Haoxiang You

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Education

Yale University Expected May 2027
PhD in Mechanical Engineering (Robotics Track), GPA: 4.0/4.0

University of Pennsylvania May 2023
MS in Mechanical Engineering (Robotics Track), GPA: 4.0/4.0

Ningbo University Jun 2020
BS in Mechanical Engineering

Purdue University Dec 2018
Visiting Student

Summary

Final-year Robotics PhD candidate at Yale with experience across the robotics stack — hardware design, control, data collection, simulation, model training, agentic workflows, and real-robot deployment. My research develops efficient robot learning algorithms. More recently, my interests lie in agentic robotics: evaluating coding agents on long-horizon physical tasks, integrating them as the robot brain, and using them as a scalable data engine for training low-level cerebellum policies such as VLAs. You can find some examples here.

Publications

EmbodiedSWE: Coding Agents for Long-Horizon Dexterous Robotics Under Review
Haoxiang You, *Z. Shen, Y. Liu, Z. Zheng, L. Zha, K. Yamazaki, M. Zhang, S. Huang, J. Sun, Q. Chen, L. He, H. Chang, D. Shah, M. Schwager, K. Fragkiadaki, P. Henderson, I. Abraham, C. Xu*

Efficient On-policy Visual-RL via Stochastic Decoupled Policy Gradient CoRL 2026
Haoxiang You, *Y. Liu, D. Zong, Qian Wang, T. Vitchutripop, Qi Wang, D. Rakita, I. Abraham*

Sample-Based Hybrid Mode Control: Asymptotically Optimal Switching of Algorithmic and Non-Differentiable Control Modes ICRA 2026
Yilang Liu, Haoxiang You, *Ian Abraham*

Goal-Driven Reward by Video Diffusion Models for Reinforcement Learning CVPR 2026
Q. Wang, M. Wu, Y. Zhang, M. Yuan, W. Zhang, Haoxiang You, *Y. Wang, X. Jin, X. Yang, W. Zeng*

Accelerating Visual-Policy Learning through Parallel Differentiable Simulation NeurIPS 2025 **[Spotlight]**
Haoxiang You, *Yilang Liu, Ian Abraham*

Is Bellman Equation Enough for Learning Control? Preprint 2025
Haoxiang You, *Lekan Molu, Ian Abraham*

Work Experience

Genesis-AI, Research Scientist Intern Jul 2025 – Nov 2025

– Built exoskeleton teleoperation hardware and a data-collection pipeline that gathers robot demonstrations for Vision-Language-Action (VLA) and imitation-learning training, and constructed digital twins of real scenes for simulation.

– Developed a general-purpose Model Predictive Control (MPC) framework that generates manipulation data autonomously, without human teleoperation, and contributed training optimizations to the sim-to-real co-training pipeline.

University of Pennsylvania, Robotics Research Assistant May 2022 – Sep 2022

– Developed control strategies — trajectory optimization, feedback control, system identification, and state estimation — for the Cassie bipedal robot, and transferred locomotion controllers from simulation to hardware, bridging the sim-to-real gap under model mismatch, sensor noise, and actuator dynamics.

YITU Tech, Self-Driving Control Intern Jan 2021 – Jul 2021

– Trained neural vehicle-dynamics models to improve simulator fidelity for self-driving control, and tuned trajectory-tracking controllers on electric test vehicles.

Selected Projects

Yale MCube Satellite 2024 – 2025

– Led the mechanical team — redesigning the satellite's structure for manufacturing — and developed communication and control code as a core member of the embedded team.

Full-Stack LLM Implementation 2025

– Implemented an LLM training and deployment stack end-to-end — pretraining, supervised fine-tuning, reinforcement learning, and distributed training.

Invited Talks & Media

Moonshot AI (Kimi) Invited Talk, From Coding Agents to Embodied Intelligence 2026
NVIDIA Robotics Seminar, Computational Methods for Curious Robots 2025
Robot Mind Podcast, Accelerating Reinforcement Learning with Visual Inputs 2025

Teaching & Academic Service

Reviewer, ICRA, CVPR, ECCV, NeurIPS, CoRL, RA-L
Organizer, Yale Robotics Reading Group
Teaching, Mechatronics Laboratory & Multivariable Calculus (Yale); Learning in Robotics (Penn)

Honors and Awards

Outstanding Academic Award	University of Pennsylvania	2023
Dean's List	Purdue University	2018
Province Scholarship	Zhejiang	2017 & 2018
First-class Academic Scholarship	Ningbo University	2017

Skills

Programming	Python, C, C++, C#
Physical Simulation	MuJoCo, Brax, Isaac Sim, Newton, Genesis, ManiSkill, Drake
ML & Optimization	PyTorch, JAX, SNOPT, IPOPT
LLM Agents	Agent harnesses & evaluation, agentic coding workflows
Hardware & Tools	SolidWorks, mechatronics, 3D printing, CNC machining, Unity, Blender